Higher National Diploma in Information Technology

Second Year first Semester Examination-2021

HNDIT 3042 Database Management System

Question 1 (Total 20 marks)

1. Define the terms of Database and Database Application Programs. (2 marks)

2. List two advantages and two disadvantages of DBMSs. (4 marks)

3. State four types of people who participate in the DBMS environment. (4 marks)

4. Draw the stages of the Database System Development Lifecycle. (5 marks)

5. Briefly explain the ANSI-SPARC three level architecture. (5 marks)

Question 2 (Total 20 marks)

1. Consider the following information about ABC hospital

• ABC Hospital needs to create a database to store information about its patients, doctors, wards and each patient's medical record.

• The personal details of every patient include patient number, name, a mobile number and the permanent address (home number, street, city).

• Doctors have doctor id, name and several phone numbers.

• In this hospital, doctors have two types of positions: ward in-charge doctors and

consultant doctors.

• Each consultant doctor has specialist field

• Age and years of experience must be recorded for ward in-charge doctor.

• Each word has unique ward number, name, ward in-charge doctor and patients.

• Consultant doctor recommends a ward 10 admit lie patter

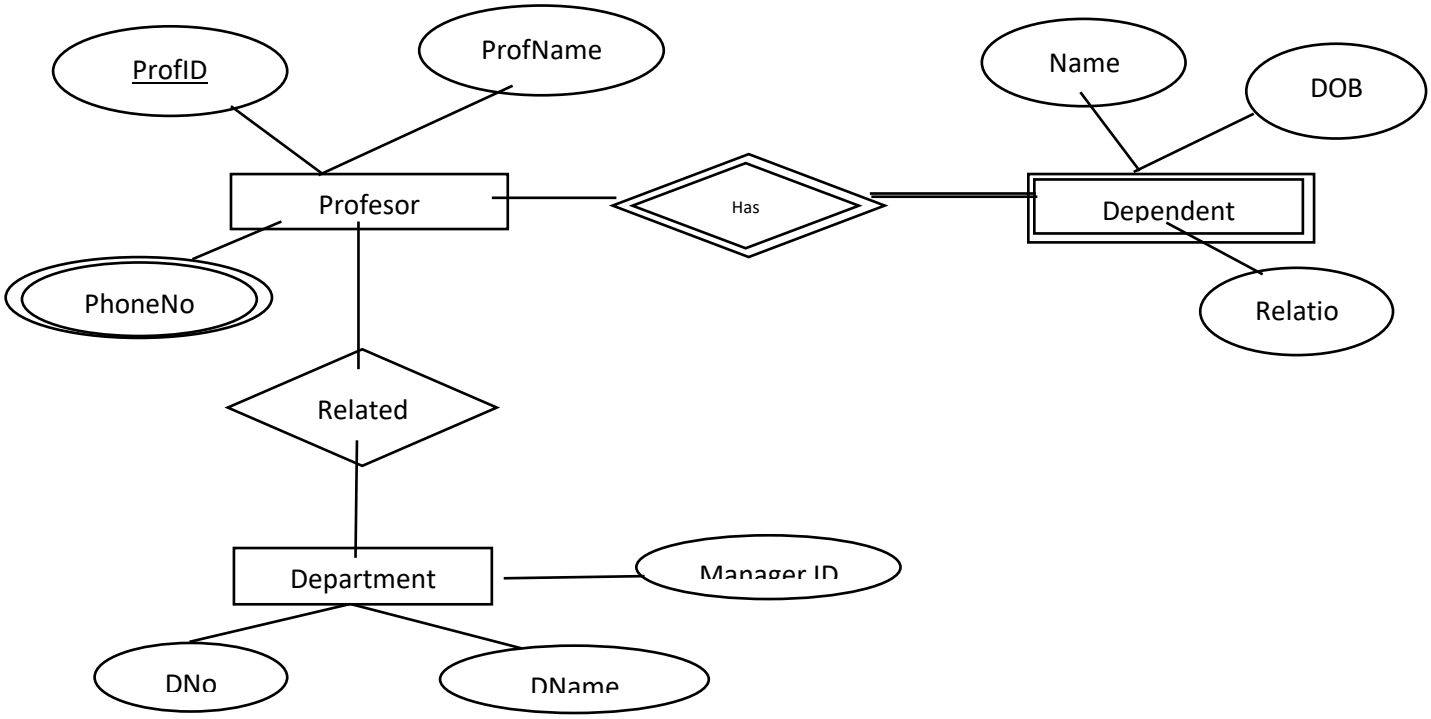
• Each patient has a medical record. It includes patient number, record id, date of the

Medical examination and discase problem.

Draw an entity Relationship (ER) Enhanced Entity Relationship (EER) Diagram to represent the above information.
indicate primary keys, composite attribute, multivalued attribute, Week entity (with partial key and identifying relationship), “ISA” relationship and cardinality constraints.

(12 Marks)

2. convert the bellow ER Diagram in to a Relational Schema. Be certain to Indicate primary keys and referential constraints.



Question 3

1. A library uses the following table to store the details of its member, the books they have borrowed and the author details. Primary key of the table id MID.

MID	Name	ISBN	BookName	Author ID	AuthorName	BorrowedDate
M01	R.R. Silva	1396	Madol Doowa	A01	M.Wickramasinghe	2018/01/06
		2009	Amba Yaluwo	A02	T.B.Illangarathne	2020/05/09
		8000	Senkottan	A006	M.P.Masimbula	2021/05/09
M05	K.Kumari	2903	Sakkaran	A0200	C.Welagedara	2017/05/20
		1396	Madol Doowa	A01	M.Wickramasinghe	2019/05/20
M09	Y.Perera	2009	Amba Yaluwo	A02	T.B.Illangarathne	2022/08/03

Note: MID stands for Member ID, Mname Stand for Member Name, ISBN Stands for Book Number.

a. Write down the table that are in Unnormalized form. (1 Mark)
b. Identify the Repeating group of attributes and transform the above Unnormalized table into tables that are in 1st Normal form. (2 Marks)
c. Identify the partial dependencies and transform above 1st Normal form table into tables that are in 2nd Normal form. (3 Marks)
d. Identify any transitive dependencies and transform above 2nd normal form table into tables that are in 3rd Normal form. (4 Marks)

2. consider the following relational schema.

Student (StudentID, FName, LName, GPA, ProjectNo)

Project (ProjectNo, PName, Budget)

Write the relational algebra queries for following questions.

a) List the Student IDs and GPAs of all the students (1 Mark)

b) List all the Project Names with a budget greater than Rs. 10000 (1 mark)

c) List all the FName, LName and rename it as "FirstName", "LastName" respectively. Also rename the relation as "Result". (2 marks)

d) List all the IDs of Students who work on the project no "POOl" and have GPA greater than 3.0. (3 marks)

e) List all the Names (FName and LName) of Students who work on a project with a budget greater than or equal Rs.50000. (3 marks)

Question 4 (Total 20 marks)

1. Write down SQL statements for the following questions.

a. Create the following tables. (4 marks)

Student (StudentID:CHAR(4), SName:VARCHAR(20), GPA:REAL, ProjectNo:CHAR(4))

Project (ProjectNo:CHAR(4), PName:VARCHAR(20), Budget:FLOAT)

b) Add a column "Location" (Location:VARCHAR(20)) to the Project table. (1 mark)

c) Insert the following information to the Student table. (2 marks)

StudntID	SName	GPA	ProjectNo
S001	Saman Perera	3.2	P01
S002	Nayani Gamage	3.0	P02

d) Delete the row where StudentID is "S002" in the Student table. (1 mark)

e) Update Saman Perera's GPA to 3.4. (2 marks)

2. Consider the following relational schema.

Student (<u>StudentID</u>, SName, GPA, ProjectNo)

Project (<u>ProjectNo</u>, PName, Budget)

Write down SQL queries for the following questions based on the above relational schema.

a) Display names of the students who are working on 'XYZ' project and have GPA greater than 3.5. (3 marks)

b) List all the project names which are starting with letter 'A' or 'a'. (2 marks)

c) List the name and budget of all projects, arranged in descending order of budget. (2 marks)

d) Find the number of students for each project. Display project number and number of students. (3 marks)

Question 5 (Total 20 marks)

1. What are ACID Properties of transaction. (2 marks)

2. Draw the State Transaction Diagram and describes the states in a transaction. (6 marks)

3. Define what is meant by the term 'database security' (2 marks)

4. Briefly explain the following computer-based security controls (6 mark)

   a. Views
   b. Backup
   c. Integrity

5. What are the two types of Access Controls? Explain them briefly. (4 marks)